

2. RIDGE vs LASSO REGRESSION (Using California Housing Dataset)

1. Aim

To develop Ridge and Lasso regression models using California Housing dataset, evaluate their performance and compare both methods.

2. Objective

- What is regression
- What is regularization
- Ridge regression concept
- Lasso regression concept
- Model evaluation
- Feature selection
- Model comparison

3. Theory

What is Regression?

Regression predicts continuous values.

Examples:

- House price prediction
- Salary prediction
- Temperature prediction

Linear regression predicts using equation:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots$$

Problem with Linear Regression

- Overfitting (memorizes data)
- Large coefficient values
- Complex model
- Multicollinearity

Solution → Regularization

A technique to control model complexity.

Regularization adds penalty to model.

Loss = Error + Penalty

Purpose:

- ✓ Reduce overfitting
- ✓ Reduce model complexity
- ✓ Improve performance

Ridge Regression (L2 Regularization)

- Reduces coefficient values
- Keeps all features
- Prevents overfitting

Formula idea:

Error + $\lambda \times$ sum of squared coefficients

Coefficients become small but never zero.

Lasso Regression (L1 Regularization)

- Makes some coefficients zero
- Removes unnecessary features
- Performs feature selection

Formula idea:

Error + $\lambda \times$ sum of absolute coefficients

Some features removed automatically.

4. Dataset Explanation - California Housing

What dataset predicts

House price in California.

Input Features

- MedInc → income
- HouseAge → house age
- AveRooms → average rooms
- AveBedrms → bedrooms
- Population → population
- AveOccup → occupancy
- Latitude → location
- Longitude → location

Target

MedHouseVal → house price

Continuous value → regression problem.

5. Algorithm

1. Load dataset
2. Split data into training and testing
3. Scale features
4. Train Ridge model
5. Train Lasso model
6. Evaluate performance
7. Compare coefficients
8. Draw conclusion

6. Complete Python Code

```
# Import required libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
# Load dataset
```

```
from sklearn.datasets import fetch_california_housing
```

```
# Split data
```

```
from sklearn.model_selection import train_test_split
```

```
# Feature scaling
```

```
from sklearn.preprocessing import StandardScaler
```

```
# Models
```

```
from sklearn.linear_model import Ridge, Lasso
```

```
# Evaluation metrics
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

Step 1: Load California Dataset

```
data = fetch_california_housing(as_frame=True)
```

```
df = data.frame
```

```
print("Dataset Preview:")
```

```
print(df.head())
```

Step 2: Separate Input and Output

```
X = df.drop("MedHouseVal", axis=1) # input features
```

```
y = df["MedHouseVal"]           # target variable
```

Step 3: Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y,
```

```
    test_size=0.2,
```

```
    random_state=42  
)
```

Step 4: Feature Scaling

Important for Ridge & Lasso

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

Step 5: Train Ridge Model

```
ridge = Ridge(alpha=1.0) # alpha = lambda
```

```
ridge.fit(X_train, y_train)
```

```
ridge_pred = ridge.predict(X_test)
```

Step 6: Train Lasso Model

```
lasso = Lasso(alpha=0.1)
```

```
lasso.fit(X_train, y_train)
```

```
lasso_pred = lasso.predict(X_test)
```

Step 7: Evaluate Performance

```
print("\nRidge Results")
```

```
print("MSE:", mean_squared_error(y_test, ridge_pred))
```

```
print("R2 Score:", r2_score(y_test, ridge_pred))
```

```
print("\nLasso Results")
```

```
print("MSE:", mean_squared_error(y_test, lasso_pred))
```

```
print("R2 Score:", r2_score(y_test, lasso_pred))
```

Step 8: Compare Coefficients

```
print("\nRidge Coefficients:")
```

```
print(ridge.coef_)
```

```
print("\nLasso Coefficients:")
```

```
print(lasso.coef_)
```

7. Output Interpretation

Mean Squared Error (MSE)

- Lower → better model.

R² Score

- Closer to 1 → good model.

Coefficients

- Ridge → all features used.
- Lasso → some features removed.

8. Comparison of Ridge vs Lasso

Feature	Ridge	Lasso
Penalty	L2	L1
Removes features	No	Yes
Coefficients	Small	Some zero
Model	Stable	Simple
Feature selection	No	Yes

9. Expected Observation

- Ridge reduces coefficient magnitude.
- Lasso sets some coefficients to zero.
- Both methods reduce overfitting.
- Model performance differs.

10. Result

Ridge and Lasso regression models were developed and evaluated. Ridge reduces coefficient values while Lasso performs feature selection. Both methods help reduce overfitting.

11. Conclusion

Ridge regression controls model complexity by shrinking coefficients. Lasso regression performs feature selection by making some coefficients zero. Both improve model performance.

Lab Record Format

- Aim
- Theory
- Algorithm
- Procedure
- Code
- Output
- Result
- Conclusion

